GGG-B-191a
March 1, 1963

SUPERSEDING
Fed. Spec. GGG-B-191
November 18, 1955

FEDERAL SPECIFICATION

BENDER, TUBE, HAND, (PORTABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand-operated benders for use in both precision and nonprecision cold bending of round metal tubing (pipe, conduit, and heavy steel tubing excluded) used in aircraft hydraulic systems, instrument control lines, fuel and oil lines on engines, refrigeration, plumbing, heating, and other installations.

1.1.1 *Federal specification coverage.* Federal specifications do not include all types and classes of tube benders which are commercially available, but only those which are generally used by the Federal Government.

1.2 Classification. Tube benders shall be of the following types and classes, as specified (see 6.2):

Type I.—Worm-gear actuated, multiple size.

Type II.—Pinion-gear actuated.

Class 1.—Multiple size.

Class 2.—Single size.

Type III.—Lever actuated.

Class 1.—Multiple size.

Class 2.—Single size.

Type IV.—External coil spring, hand-actuated.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

TT-C-490—Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-485—Enamel, Semi-Gloss, Rust-Inhibiting.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-E-15090—Enamel, Equipment, Light-Gray.

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Society for Testing Materials Standard:

A-269—Seamless and Welded Austenitic Stainless Steel Tubing for General Service.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of tube benders otherwise in accordance with this specification.

3.2 Material. Material used in construction of tube benders shall be sound, of uniform quality and condition to provide ample strength, and shall conform in composition, heat treatment, and suitability to preclude appreciable deformation of any part when subjected to hard usage. Tube benders shall meet the applicable test specified in 4.3.

3.3 Construction. Tube benders shall be so constructed that no assembled part will become detached while in use. All moving parts shall operate without binding and shall be constructed so that excessive looseness and play adversely affecting operation will not develop due to hard usage.

3.4 Design. Tube benders shall be of the latest acceptable design, convenient to operate, and entirely adequate for the purpose intended. All parts subject to wear and normally replaceable shall be readily accessible for adjustment or repair.

3.5 Threads. All threaded parts shall be of a nationally recognized commercial standard thread having standard size, shape, and tolerances in order to insure correct mating of parts and convenience of interchangeability and replacement.

3.6 Lubrication. All bearings and surfaces requiring periodic lubrication shall be provided with accessible means for lubricating.

3.7 Surface finish. All castings, forgings, or welded parts shall be thoroughly cleaned and shall be free of sand, dirt, fins, sprues, scale, flux, or other harmful or extraneous materials. All external surfaces and edges shall be free of burrs, fins, and sharp edges.

3.8 Corrosion-resisting coatings. Type IV benders and material in components of all other types of benders which is subject to corrosion, shall be coated with either zinc, cadmium, hot phosphate, or oxide in accordance with good commercial practice.

3.9 Interchangeability of parts. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:38Z
Check the source to verify that this is the current version before use.

3.10 Operating instructions. All tube benders, except type IV, shall be furnished with two sets of printed instructions covering operation and maintenance. When carrying cases are furnished, the instructions shall be enclosed within; otherwise, they shall be attached to the bender.

3.11 Identification marking. All tube benders, except type IV, shall be marked in a plain and permanent manner with the manufacturer's part or model number and the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. For military purchases, marking shall be in accordance with MIL-STD-130, as practicable.

3.12 Type I, worm-gear actuated, multiple size. Type I tube benders shall be designed for making precision bends on cold-drawn, corrosion-resisting steel tubing conforming to AISI type 304 of ASTM Standard A-269 without the aid of mandrels.

3.12.1 *Operation and functional design.* The operating mechanism of type I benders shall have sufficient mechanical advantage to provide a smooth, continuous operation in producing bends of approximately 180 degrees on tubing of the diameter shown in table I without subjecting an operator of average strength to undue strain or fatigue. The design shall be such that after making a bend of approximately 180 degrees the tubing can be easily removed from the bender. Means shall be provided for clamping the bender in a standard machinist's bench vise having a jaw opening of 4 inches or more.

3.12.2 *Overload pin or safety device.* Each tube bender shall be provided with an overload pin or safety device in the event failure occurs due to overloading. An attempt to bend heavy-walled tubing beyond the capacity of the bender would only result in shearing the overload pin which could be readily replaced and prevent serious damage to the bender.

3.12.3 *Angle of bends.* The bender shall be capable of bending the specified tubing to any angle from 0 to approximately 180 degrees without having to reset or adjust the tubing. Clearly visible means shall be provided on the bender for measuring angles of bend in 15-degree increments up to approximately 180 degrees. Benders shall be capable of being positioned at any point on the tubing and also making various styles of bends such as offset, return, and right angle.

3.12.4 *Quality of bends.* The bender shall be capable of accomplishing the desired bends without marring the bent section with diemarks, kinks, wrinkles, or surface imperfections. The maximum reduction in cross-

Table I - Type I tube bender, worm-gear actuated, multiple size.

Nominal outside diameter of tubing	Range of the nominal wall thickness of the tubing	Radius of bend measured to centerline of the tubing
		Plus 1/16 or minus 1/8
Inch	Inch	Inches
1/4	0.016 through 0.049	9/16
5/16	.020 through .065	11/16
3/8	.028 through .083	15/16
1/2	.035 through .083	1- 1/4
5/8	.042 through .083	1- 1/2
3/4	.049 through .083	1- 3/4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

sectional diameter at any point on the bend shall not exceed 5 percent of the original diameter of the tubing when bends are made within the radius range shown in table I.

3.12.5 *Weight and dimensions.* The combined weight of the tube bender, accessories, and carrying case shall not exceed 40 pounds. The dimensions of the carrying case shall not exceed 16-1/8 by 8-1/2 by 7-1/8 inches, including hasps, hinges, and handles.

3.12.6 *Carrying case.* Each tube bender and its accessories shall be furnished in a metal carrying case with adequate means provided for securing loose parts within the case. The case shall be of sufficient rigidity to withstand ordinary handling throughout the normal life of the bender. The case shall have a hinged lid with at least one drawbolt or other type of positive catch and a hardy full-grip handle securely attached to the lid. Both the interior and exterior surfaces of the case shall have at least one protective coat of paint. The case shall be finished as specified in 3.16.

3.12.7 *Identification marking.* Parts used in forming the bends such as sheaves, form blocks, slide blocks, or wiping dies shall be marked in a plain and permanent manner with the outside diameter of the tubing for which the part is used.

3.12.8 Type I tube benders shall be furnished with all necessary parts for bending all nominal sizes of the tubing specified in table I. They shall conform to table I and shall be similar to figure 1.

3.13 **Type II, pinion-gear actuated.**

3.13.1 *Quality of bends.* The type II bender shall be capable of accomplishing the desired bends without kinks, wrinkles, or scratches which would likely contribute to failure of the bent section in service. The maximum reduction in cross-sectional diam-

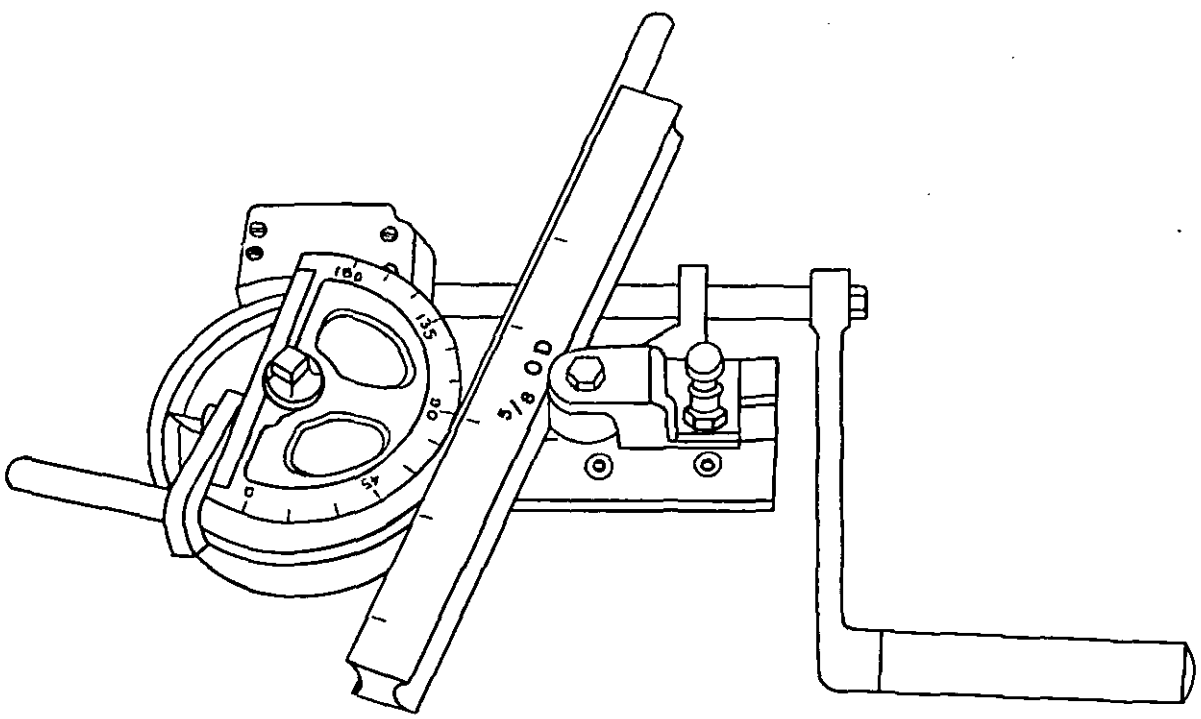


FIGURE I.—Type I, tube bender, worm-gear actuated, multiple size.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

eter at any point on the bend shall not exceed 5 percent of the original diameter of the tubing when bends are made on soft copper, aluminum, and medium-heavy steel tubing, or 10 percent when bends are made on thin-walled hard-drawn copper and steel tubing. (The term "medium-heavy steel tubing" covers the following wall thicknesses for 3/8-, 1/2-, 5.8-, 3.4-, 7/8-, 1-, and 1-1/8-inch sizes of tubing: 0.035—0.060, 0.049—0.065, 0.049—0.072, 0.065—0.083, 0.065—0.083, 0.065—0.083, and 0.065—0.083 inch, respectively.) (The term "thin-walled hard-drawn copper and steel tubing" covers the following wall thicknesses for 3/8-, 1/2-, 5/8-, 3/4-, 7/8-, 1-, and 1-1/8-inch sizes of tubing: 0.030—0.042, 0.035—0.049, 0.040—0.055, 0.042—0.060, 0.045—0.065, 0.045—0.065 and 0.045—0.065 inch, respectively.)

3.13.2 *Class 1, multiple size.* Class 1 tube benders shall be designed for making accurate bends on soft steel, aluminum, and copper tubing. When specified (6.2), the bender shall also be capable of properly bending hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy steel tubing.

3.13.2.1 *Operation and functional design.* The operating mechanism of class 1 benders shall have sufficient mechanical advantage to bend the maximum size tubing shown in table II without undue effort by the operator. The design shall be such that after making a bend of 180 degrees the tubing can be easily removed from the bender. The benders shall be so designed that they can be held by hand or clamped in a bench vise. They shall be open-side benders so they can be slipped over the tubing at the exact point where a bend is required.

3.13.2.2 *Angle of bends.* The bender shall be capable of bending tubing of the diameter shown in table II to any angle from 0 to 180 degrees without having to reset or adjust the tubing. Clearly visible means shall be provided on the bender for measuring angles of bend in 45-degree increments up to 180 degrees.

3.13.2.3 *Weight.* The combined weight of the tube bender and accessories shall not exceed 12 pounds.

3.13.2.4 *Identification marking.* Forming wheels, sheaves, or blocks shall be marked in a plain and permanent manner with the outside diameter of the tubing for which the part is designed.

3.13.2.5 Type II, class 1 tube benders shall be furnished with all necessary parts for bending all nominal sizes of the tubing listed in table II. They shall conform to table II and shall be similar to figure 2.

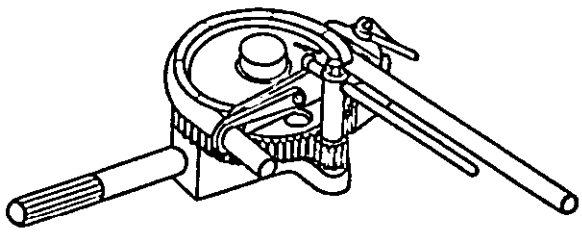


FIGURE 2.—Type II, class 1 tube bender, pinion-gear actuated, multiple size.

Table II - Type II, class 1 tube bender, pinion-gear actuated, multiple size.

Nominal outside diameter of tubing	Range of the nominal wall thickness of the tubing	Radius of bend measured to centerline of the tubing	
		Minimum	Maximum
Inch	Inch	Inches	Inches
3/8	0.028 through 0.083	15/16	1-5/16
1/2	.035 through .083	1- 1/2	2
5/8	.049 through .083	2	2-1/2
3/4	.049 through .083	2- 1/2	3-1/4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

3.13.3 *Class 2, single size.* Class 2 tube benders shall be designed for making bends on any type of tubing, except hard-drawn copper or cold-drawn steel tubing having a wall thickness of 0.050 inch or less for the 1-inch or 1-1/8-inch size. (The term "any type of tubing" covers the following wall thicknesses for 3/8-, 1/2-, 5/8-, 3/4-, 7/8-, 1-, and 1-1/8-inch sizes of tubing: 0.030—0.060, 0.035—0.065, 0.040—0.072, 0.042—0.083, 0.045—0.083, 0.065—0.095, and 0.065—0.095 inch, respectively.)

3.13.3.1 *Operation and functional design.* The operating mechanism of a class 2 bender shall have sufficient mechanical advantage to bend any type of tubing in the size specified without undue effort by the operator. The benders shall be so designed that they can be held by hand, bolted to a bench, or clamped in a bench vise. They shall be of heavy-duty design and open-side type so they they can be positioned on the tubing at any point where a bend is required.

3.13.3.2 *Angle of bends.* Each bender shall be capable of bending tubing to any angle from 0 to 90 degrees without having to reset or adjust the tubing. Clearly visible means shall be provided on each bender for measuring angles of bend at 15, 30, 45, 60, and 90 degrees. Each bender shall be capable of making both right- and left-hand bends as well as offset, return, and right-angle bends. The smallest coil or return bend that each bender will make shall be no larger than the applicable dimension listed in table III.

3.13.3.3 *Identification marking.* In addition to the marking specified in 3.11, each bender shall be marked in a plain and permanent manner with the outside diameter of the tubing for which the bender is designed.

3.13.3.4 Type II, class 2 tube benders shall conform to the requirements shown in table III for the size specified (6.2), and shall be similar to figure 3.

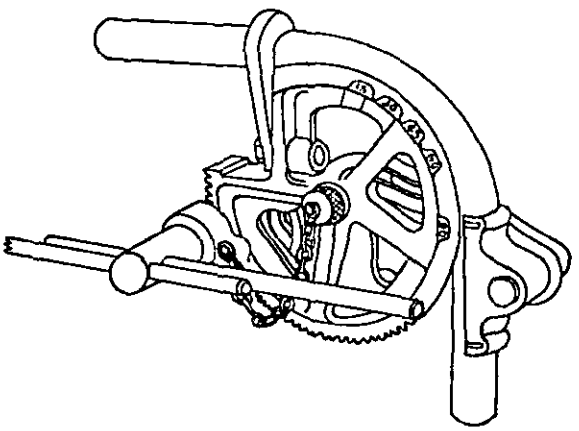


FIGURE 3.—Type II, class 2 tube bender, pinion-gear actuated, single size.

Table III - Type II, class 2 tube bender, pinion-gear actuated, single size.

Bender size and nominal outside diameter of tubing	Coil bend or return bend measured from centerline to centerline of tubing (see 3.13.3.2)	Radius of bend measured to centerline of the tubing		Weight
		Minimum	Maximum	Maximum
Inches	Inches	Inches	Inches	Pounds
3/8	4-1/2	1-5/8	1-7/8	3-1/4
1/2	6	2-3/8	2-5/8	5
5/8	7-1/2	2-7/8	3-1/8	9-1/2
3/4	9-1/2	3-3/4	4-1/4	15-1/4
7/8	9-1/2	3-3/4	4-1/4	15
1	9-1/2	3-3/4	4-1/4	15
1-1/8	9-1/2	3-3/4	4-1/4	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

3.14 Type III, level actuated.

3.14.1 *Class 1, multiple size.* Class 1 tube benders shall be designed for making accurate bends on soft-copper, aluminum, steel, or other soft-metal tubing. When specified (6.2) the bender shall also be capable of properly bending hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy steel tubing. The maximum wall thicknesses for 1/4-, 5/16-, 3/8-, 1/2-, 5/8-, 3/4-, and 7/8-inch sizes of tubing which class 1 benders will bend without deformation of the bender and without undue effort by the operator are as follows: 0.049, 0.049, 0.055, 0.060, 0.060, 0.065 and 0.065 inch, respectively.

3.14.1.1 *Operational and functional design.* The benders shall have sufficient leverage to maintain a smooth performance throughout the bending operation. They shall be of the open-side type and capable of being positioned on the tubing at any point where a bend is required. The benders shall have a removable handle and be so designed that the tubing can be easily freed after bending. They shall also be designed for medium heavy-duty performance with provisions made for clamping in a bench vise.

3.14.1.2 *Angle of bends.* Each bender shall be capable of making the angles of bend specified in 3.13.2.2.

Table IV - Type III, class 1 tube bender, lever actuated, multiple size.

Bender size Number	Tubing capacity, nominal outside diameter of the tubing	Minimum wall thickness without exceeding 5 percent reduction of cross-sectional diameter	Radius of bend measured to centerline of the tubing		Weight of bender complete (includes carrying case when furnished) (see 3.14.1.4)
			Minimum	Maximum	Maximum
	Inch	Inch	Inches	Inches	Pounds
1	1/2	0.035	1-1/2	2-1/8	9-1/2
	5/8	.035	2-1/8	2-5/8	
2	3/8	.035	1-1/2	1-3/4	20
	1/2	.035	1-1/2	1-3/4	
	5/8	.035	2-1/8	2-3/8	
	3/4	.035	2-7/8	3-1/8	
3	1/4	.028	5/8	7/8	20-1/2
	5/16	.028	7/8	1-1/8	
	3/8	.035	1-1/8	1-3/8	
	1/2	.035	1-1/2	1-3/4	
	5/8	.035	2-1/8	2-3/8	
	3/4	.035	2-7/8	3-1/8	
4	1/4	.028	5/8	7/8	26-1/2
	5/16	.028	7/8	1-1/8	
	3/8	.035	1-1/8	1-3/8	
	1/2	.035	1-1/2	1-3/4	
	5/8	.035	2-1/8	2-3/8	
	3/4	.035	2-7/8	3-1/8	
	7/8	.042	2-7/8	3-1/8	
5	5/8	.035	2-1/8	2-5/8	10
	7/8	.042	2-7/8	3-1/8	
6	3/4	.035	2-7/8	3-1/8	11
	7/8	.042	2-7/8	3-1/8	
7	5/8	.035	2-1/8	2-5/8	10
	3/4	.035	2-7/8	3-1/8	

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.

3.14.1.3 *Quality of bends.* The bender shall be capable of accomplishing the desired bends without kinks, wrinkles, or scratches which would likely contribute to failure of the bent section in service. The maximum reduction in cross-sectional diameter at any point on the bend shall not exceed 5 percent of the original diameter of the tubing when bends are made on soft tubing having a wall thickness equal to or greater than that specified in table IV.

3.14.1.4 *Carrying case.* A carrying case shall be furnished with each size number 2, 3, and 4 tube bender. The case shall conform to 3.12.6, and the maximum dimensions shall be 20-1/2 by 6-1/2 by 6-3/8 inches. The case shall be finished as specified in 3.16.

3.14.1.5 *Identification marking.* Forming wheels, sheaves, or blocks shall be marked in a plain and permanent manner with the outside diameter of the tubing for which the part is designed.

3.14.1.6 Type III, class 1 benders shall be furnished with all necessary parts for bending the applicable sizes of tubing listed for that particular bender size number in table IV, and shall otherwise conform to table IV and be similar to figure 4.

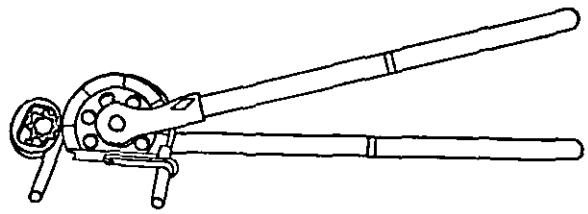


FIGURE 4.—Type III, class 1 tube bender, lever-actuated, multiple size.

3.14.2 *Class 2, single size.* Class 2 tube benders shall be designed for making accurate bends on soft-copper, aluminum, or steel tubing, except that the 1-inch size is not recommended for steel tubing. When specified (6.2) benders in sizes 1/8 inch through 7/8 inch shall be capable of properly bending hard-drawn copper, annealed corrosion-resisting steel, and medium heavy steel tubing. The maximum wall thicknesses for 1/8-, 3/16-, 1/4-, 5/16-, 3/8-, 7/16-, 1/2-, 5/8-, 3/4-, 7/8- and 1-inch sizes of tubing which class 2 benders will bend without deformation of the bender and without undue effort by the operator are as follows: 0.042, 0.049, 0.049, 0 .049, 0.055, 0.055, 0.060, 0.060, 0.065, 0.065, and 0.083 inch, respectively.

3.14.2.1 *Operation and functional design.* The benders shall have sufficient leverage

TABLE V.— *Type III, class 2 tube bender, lever actuated, single size*

Bender size and nominal outside diameter of tubing	Minimum wall thickness without exceeding 5 percent reduction of cross-sectional diameter	Radius of bend measured to centerline of the tubing		Weight (maximum)
		Minimum	Maximum	
Inch	Inch	Inches	Inches	Pounds
1/8	0.028	3/8	7/16	5/8
3/16	.028	7/16	1/2	5/8
1/4	.032	9/16	5/8	4
5/16	.032	11/16	13/16	4-3/4
3/8	.035	7/8	1	5-1/2
7/16	.042	1	1- 1/2	7
1/2	.042	1- 1/4	1- 1/2	7
5/8	.042	1- 1/2	2- 1/4	10
3/4	.049	1- 3/4	3	16
7/8	.049	2	3	19
1	.065	2- 1/2	3- 1/2	32

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

to maintain a smooth performance throughout the bending operation. They shall be capable of producing bends of 180 degrees and of being positioned on the tubing at any point where a bend is required. The benders for tubing of 5/8-inch-outside diameter and larger shall be designed so that they may be clamped in a bench vise.

3.14.2.2 *Angle of bends.* The benders shall be capable of bending tubing to any angle from 0 to 180 degrees without having to reset or adjust the tubing. Clearly visible means shall be provided on the bender for measuring angles of bends.

3.14.2.3 *Quality of bends.* The bender shall be capable of accomplishing bends in conformance with 3.14.1.3 except that the wall thickness of tubing shall be equal to or greater than that shown in table V.

3.14.2.4 *Identification marking.* Each tube bender shall be marked in a plain and permanent manner with the outside diameter of the tubing for which the bender is designed.

3.14.2.5 Type III, class 2 tube benders shall conform to the requirements shown in table V, and shall be similar to figure 5 (see figure 5a, 5b or 5c).

3.15 Type IV, external coil spring, hand-actuated. Type IV tube benders shall be designed for use in making nonprecision bends on soft-copper and aluminum tubing.

3.15.1 *Material.* The benders shall be of hard-drawn spring wire, oil-tempered, or piano wire and shall be strain relieved after coiling. The diameter of the wire shall be that which is regularly used for standard tube benders of this type.

3.15.2 *Operation and functional design.* At least one end of each bender shall be flared or other suitable means provided in order that the bender may be easily removed from the tubing after a bend is made. The coil spring shall have sufficient clearance to slide over the tubing easily. The minimum and maximum wall thicknesses for 1/4-, 5/16-, 3/8-, 7/16-, 1/2-, and 5/8-inch sizes of tubing which the coil spring bender will support sufficiently to prevent collapse of the tubing when a bend is made, are as follows: 0.030—0.035, 0.032—0.035, 0.032—0.035, 0.032—0.035, 0.032—0.035, and 0.035, —0.040 inch, respectively. The design of the bender shall be such that soft tubing will not be scratched.

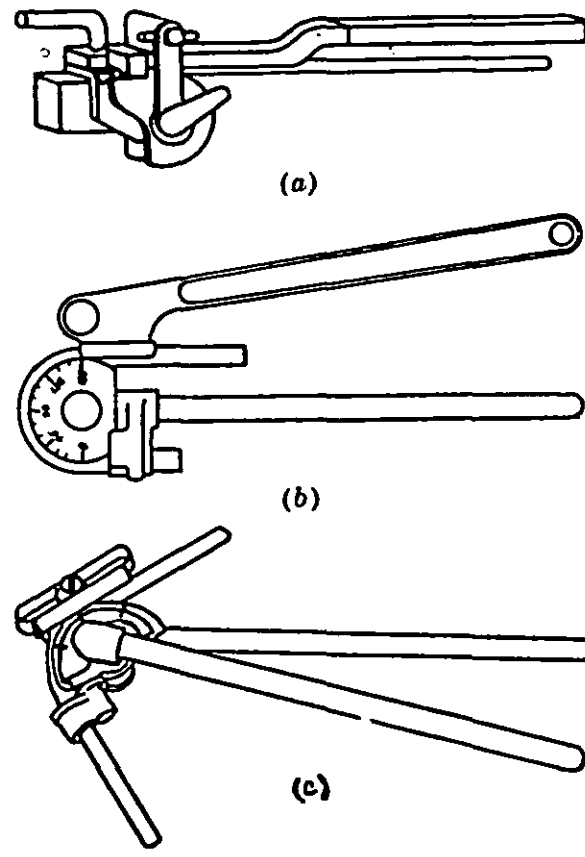


FIGURE 5.—Type III, class 2 tube bender, lever-actuated, single size.

3.15.3 *Sets.* Type IV benders shall be furnished in individual sizes or in sets of 4 or 6 sizes, as specified (6.2). Sets of 4 shall consist of sizes 1/4-, 3/8-, 1/2-, and 5/8-inch. Sets of 6 shall consist of all sizes shown in table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Table VI - Type IV tube bender, external coil spring, hand actuated.

Bender size and nominal outside diameter of tubing	Overall length		Weight	
	Minimum	Maximum	Minimum	Maximum
Inch	Inches	Inches	Ounces	Ounces
1/4	10	11	2-1/2	3-1/2
5/16	10	11	2-1/2	3-1/2
3/8	10	11	3-1/2	4-1/2
7/16	12	13	6	7
1/2	12	13	6	7-1/2
5/8	12	13	7-1/2	9

3.15.4 *Chest.* When specified (6.2), sets of 4 or 6 benders shall be furnished in a sturdily commercially constructed wood or steel chest having a hinged lid and a fastener. The chest shall be finished as specified in 3.16.

3.15.5 Type IV tube benders shall conform to table VI, and shall be similar to figure 6.

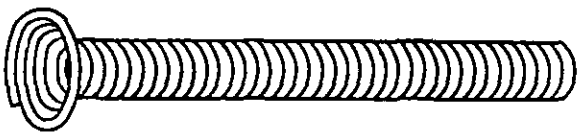


FIGURE 6.—Type IV tube bender, external coil spring, hand actuated.

3.16 Finish.

3.16.1 *Protective coating.* Unless otherwise specified (6.2), the finish and color of paint used on the interior and exterior surfaces of the carrying cases or chests shall be in accordance with manufacturer's commercial practice. When specified (6.2), the cases or chests shall be furnished with either olive drab or gray enamel as specified in 3.16.2 or 3.16.3.

3.16.2 *Olive drab.* When olive-drab finish is specified, the interior and exterior surfaces of the cases or chests shall be prepared for painting in accordance with any method of TT-C-490 and then shall be coated with enamel conforming to TT-E-485 and shall match color No. 24087 of Fed. Std. No. 595.

3.16.3 *Gray enamel.* When gray finish is specified, the interior and exterior of the cases or chests shall be coated with light gray equipment enamel in accordance with formula 111, type III, class 2 of MIL-E-15090.

3.17 Workmanship. Edges and corners of tube benders shall be rounded in conformance with the best commercial practice. Benders shall have no burrs, fins, sharp projections, excessive surface roughness, or any imperfections which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:38Z
Check the source to verify that this is the current version before use.

4.2 Inspection.

4.2.1 *Lot.* All tube benders of the same type, class and size shall be considered a lot for purposes of inspection.

4.2.2 *Examination.* A random sample of tube benders shall be selected from each lot in accordance with MIL-STD-105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300, and inspection level I for lots of 301 and over. The acceptable quality level shall be 1.5 percent defective. The benders shall be examined to verify compliance with this specification. Defects shall be classified in accordance with table VII. One or more defects in any tube bender in the sample shall cause rejection of the tube bender, and if the number of defective tube benders in any sample exceeds the acceptance number for that sample, this shall cause rejection of the lot.

4.2.3 *Tests.* A random sample of tube benders shall be selected from each lot in accordance with MIL-STD-105 at inspection level L-7. The acceptable quality level shall be equal to 6.5 percent defective; however, all the sample sizes shall be used, and the acceptance number shall be zero until the acceptable quality level permits one or greater. Each sample tube bender shall be subjected to the applicable test of 4.3 to verify compliance with this specification. Any tube bender failing to pass the applicable test shall cause rejection of the tube bender, and if the number of defective tube benders in the sample exceeds the acceptance number for that sample, this shall cause rejection of the lot.

Table VII.—Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type not as specified.
102	Class not as specified (types II and III only).
103	Size (type II, class 2, types III and IV only); if type IV sizes in set, as applicable, not as specified.
104	Material used in construction not suitable for intended use, not sound and uniform, not clean, not free from sand, dirt, fins, sprues, scale, flux or other injurious defects (types I, II, and III only).
105	Construction and design nonconforming; not actuated as specified, worm and worm gear fail to mesh properly; bender not convenient to operate (type I only).
106	Bender incomplete, component parts missing, damaged, improperly formed or positioned; parts not suitably secured to preclude becoming detached or development of excessive looseness or play during hard usage (types I, II, and III only).
107	Moving parts fail to operate without binding or excessive looseness and play (types I, II, and III only).
108	Worm or worm gear teeth damaged, cracked, broken, chipped, or improperly finished (type I only).
109	Parts subject to wear and normal replacement not readily accessible for adjustment or repair (types I, II, and III only).
110	Accessible means for lubricating not provided for all bearings and surfaces requiring lubrication (types I, II, and III only).
111	Tubing cannot be easily removed from bender after making the required maximum bend (types I, II, and III only).
112	Suitable means not provided for clamping the bender in a standard machinists' bench vise having the designated minimum jaw opening (type I only).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Table VII.—Classification of defects in accordance with MIL-STD-105 (cont'd)

Categories	Defects
Major (cont'd.):	
113	Bender not provided with an overload pin or safety device for adequate protection against overloading; overload pin, when sheared in bending tubing beyond the bender capacity, not readily replaceable (type I only).
114	Bender not capable of bending the applicable tubing to any angle within the stipulated range without having to reset or adjust the tubing (type I only).
115	Clearly visible means not provided on bender for measuring angles of bend in the designated increments up to the specified maximum angle (types I, II, and III only).
116	Bender not capable of being positioned *at any point on the tubing or making* the various stipulated styles of bends (type I only).
117	All necessary parts for bending each of the specified nominal sizes of tubing not furnished (type I; type II, class 1; and type III, class 1 only).
118	Carrying case not furnished; not suitably made of metal with sufficient rigidity to withstand the required handling; lid not hinged with a positive catch and a hardy full grip handle securely attached; adequate means not provided for securing loose parts within the case (when required) (types I and III only).
119	Dimensions of carrying case exceed the stipulated maximum limits, when applicable (types I and III only).
120	Threads nonconforming, stripped, crossed, chipped or torn (types I, II, and III only).
121	Component parts subject to corrosion not coated as required, coating not smooth, uniform, and adherent (types I, II, and III only).
122	Interior and exterior surfaces of carrying case not painted as required (when applicable) (types I and III only).
123	Replacement parts not dimensionally and functionally interchangeable without requiring modification (types I, II, and III only).
124	Parts used in forming bends not marked in a plain and permanent manner with the applicable outside diameter of the tubing for which the part is used (type I; type II, class 1; and type III, class 1 only).
125	Working surfaces of sheaves, form blocks, slide blocks, or wiping dies not smoothly finished or suitably formed, not free from galling, chatter, toolmarks, gouges, nicks, or dents (type I only).
126	External surfaces and edges not free from burrs, fins, and sharp edges (type I only).
127	Printed instructions covering operation and maintenance not inclosed in carrying case, less than the required number of sets furnished (type I only).
128	Construction and design nonconforming; not actuated as specified, pinion and gear fail to mesh properly; bender not convenient to operate (type II only).
129	Teeth of gears damaged, cracked, broken, chipped, or improperly finished (type II only).
130	Bender not capable of being held by hand or clamped in a bench vise (type II, class 1 only).
131	Bender not capable of being held by hand, bolted to a bench, or clamped in a bench vise (type II, class 2 only).
132	Bender not open-side type, cannot be slipped over tubing or positioned on tubing at the exact point where a bend is required (type II only).

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:38Z
Check the source to verify that this is the currentversion before use.

TABLE VII.—Classification of defects in accordance with MIL-STD-105 (cont'd)

Categories	Defects
Major: (cont'd.):	
133	Bender not capable of bending the applicable tubing at any angle within the stipulated range without having to reset or adjust the tubing (type II only).
134	Bender not capable of making both right and left-handed bends as well as offset, return, and right-angle bends (type II, class 2 only).
135	Bender not marked in a plain and permanent manner with the applicable outside diameter of the tubing for which the bender is designed (class 2 of types II and III only).
136	Working surfaces of forming wheel(s), sheave(s), or blocks(s) not smoothly finished or suitably formed, not free from galling, chatter, toolmarks, gouges, nicks, or dents (types II and III only).
137	Printed instructions covering operation and maintenance missing, or less than the required number of sets furnished (type II only).
138	Construction and design nonconforming; not actuated as specified; bender not convenient to operate (type III only).
139	Lever handles not removable (type III, class 1 only).
140	Bender not open-side type, cannot be positioned on tubing at any point where a bend is required (type III, only).
141	Provision not made for clamping in a bench vise (when applicable) (type III only).
142	Bender not capable of bending the applicable tubing to any angle within the stipulated range without having to reset or adjust the tubing (type III only).
143	Printed instructions covering operation and maintenance not inclosed in carrying case or attached to bender, as applicable, less than the required number of sets furnished (type III only).
144	Material nonconforming, not spring temper, not smooth and uniform, not free from cracks, splits, die-marks, burrs, fins, sharp projections, or excessive surface roughness (type IV only).
145	Wire diameter not suitable, not uniformly coil wound, deformed or not shaped true and round (type IV only).
146	End not properly flared, edges and corners not suitably rounded (type IV only).
147	Inside diameter of the coil spring fails to have the proper clearance for the applicable diameter of tubing (type IV only).
148	Overall length not within the required limits (type IV only).
149	Chest not furnished for set, when specified, material and construction not suitable; fastener not provided for lid or lid not hinged; not properly finished and painted (type IV only).
Minor:	
201	Marking, manufacturer's part or model number and manufacturer's name or trademark missing, incorrect, illegible or not permanent (types I, II, and III only).
202	Preservation, packaging, and packing not as required.

4.3 Test procedures. Filling of the tubing with sand, sulfur, or other material will not be permitted in performing the following bend tests:

4.3.1 *Test for type I tube benders.* Cold-drawn corrosion-resisting steel tubing conforming to AISI type 304 of ASTM Standard A-269 shall be used for this test. At least two bends of approximately 180 degrees shall be made on each of the six outside diameters of tubing specified for the

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.

bender, one each of the minimum and maximum wall thicknesses as specified in table I, making a total of 12 bends to be made with each bender. Performance of type I benders shall show no evidence of permanent deformation, binding of mechanism, excessive pressure required for operation, excessive looseness adversely affecting operation, any assembled part becoming detached, difficulty of adjusting or fitting replaceable parts, difficulty in removing the bent tubing, failure to conform to quality of bends specified in 3.12.4, or failure of bends to be within the range of radii specified in table I.

4.3.2 *Test for type II, class* 1 *tube benders.* Commercial copper tubing shall be used for this test. At least two bends of 180 degrees shall be made on each of the four outside diameters of tubing specified for the bender, one each of the minimum and maximum wall thicknesses as specified in table II, making a total of eight bends to be made with each bender. When the contract or order specifies that the bender will be required to bend hard-drawn copper annealed corrosion-resisting steel, and medium-heavy steel tubing (3.13.2), at least one additional bend shall be made on each of the four outside diameters specified for the bender, using hard-drawn copper of the maximum wall thickness specified in table II. The quality of bends shall conform to 3.13.1 and performance shall conform to 4.3.1. Radius of bends shall conform to table II.

4.3.3 *Test for type II, class* 2 *tube benders.* Commercial steel tubing of the minimum and maximum wall thicknesses specified in 3.13.3 shall be used for this test. At least two bends of 90 degrees shall be made on each of the tube benders, one each of the minimum and maximum wall thicknesses specified. The quality of bends shall conform to 3.13.1 and performance shall conform to 4.3.1. Radius of bends shall conform to table III.

4.3.4 *Test for type III, class* 1 *tube benders.* Commercial copper tubing shall be used for this test. At least two bends of 180 degrees shall be made on each of the outside diameters of tubing specified for the applicable size bender, one each of the minimum and maximum wall thicknesses specified in 3.14.1 and table IV. When the contract or order specifies that the bender will be required to bend hard-drawn copper, annealed corrosion-resisting steel, and medium heavy steel tubing (3.14.1), at least one additional bend shall be made on each of the outside diameters specified for the applicable size bender, using hard-drawn copper with a wall thickness at least 20 percent greater than the minimum specified in table IV. The quality of bends shall conform to 3.14.1.3 and performance shall conform to 4.3.1. Radius of bends shall conform to table IV.

4.3.5 *Test for type III, class* 2 *tube benders.* At least two bends of 180 degrees shall be made with each of the benders, using soft commercial copper tubing, one each of the minimum and maximum wall thickesses specified in 3.14.2 and table V. The quality of bends shall conform to 3.14.2.3 and performance shall conform to 4.3.1. Radius of bends shall conform to table V. When the contract or order specifies that the bender will be required to bend hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy duty steel tubing (3.14.2), at least one additional bend shall be made on each size tube specified for the applicable size bender (table V) using hard-drawn copper with a thickness at least 20 percent greater than the minimum specified in table V.

4.3.6 *Tests for type IV tube benders.* The tube benders shall not break when held firmly in a testing device shown on figure 7, and stretched twice the distance between the holding jaws for the 1/4-, 5/16- and 3/8-inch sizes, and two and one-half times for the 7/16-, 1/2- and 5/8-inch sizes. After releasing, the overall length shall not exceed the original by more than 3/16 inch. Each bender shall then be used to make a bend

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

of 180 degrees around a mandrel of the size shown on figure 8 for the applicable size soft copper tubing. The wall thickness of the copper tubing shall be within the range specified in 3.15.2. Removal of the bender shall not cause the bender to break or exceed the original length more than 1/8 inch in addition to the 3/16 inch allowed for the pull test.

4.4 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the tube benders shall be examined to determine compliance with the requirements of section 5 herein.

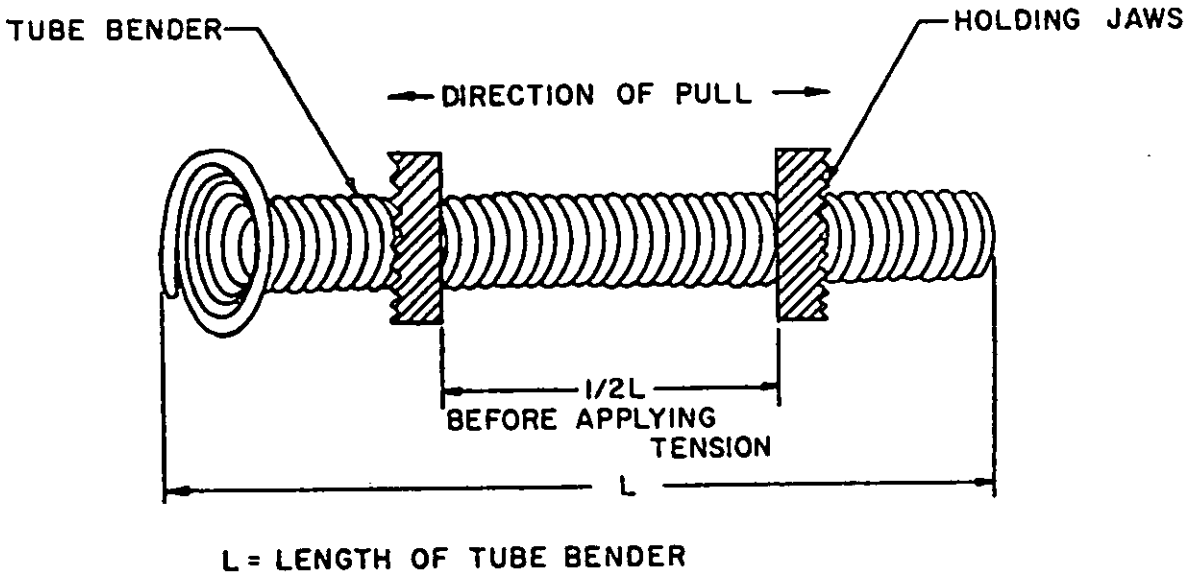


FIGURE 7.—Pull-test diagram.

MANDREL DIAMETER = D
MAXIMUM BEND RADIUS = R

TUBING O.D.	R	D
1/4	3/4	1-1/2
5/16	15/16	1-7/8
3/8	1-1/8	2-1/4
7/16	1-5/16	2-5/8
1/2	1-1/2	3
5/8	1-7/8	3-3/4

DIMENSIONS IN INCHES

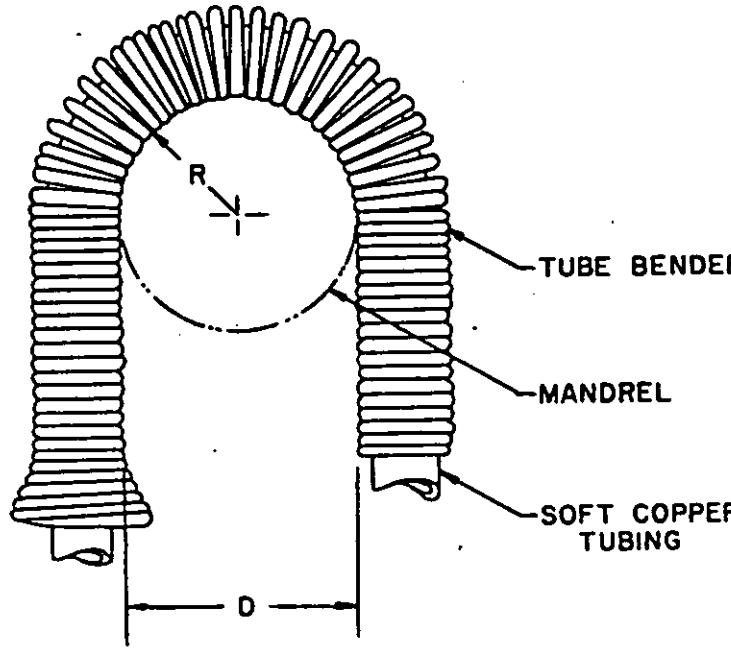


FIGURE 8.—Bend-test diagram.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:38Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* Cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Cleaning, preservation, and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Benders, packaged as specified (6.2), shall be packed in accordance with level A requirements of MIL-H-15424.

5.2.2 *Level B.* Benders, packaged as specified (6.2), shall be packed in accordance with level B requirements of MIL-H-15424.

5.2.3 *Level C.* Benders, packaged as specified (6.2), shall be packed in containers that will insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior and exterior packages shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, or herein, unit and intermediate packages and exterior shipping containers shall be marked in accordance with MIL-H-15424.

6. NOTES

6.1 Intended use.

6.1.1 *Type I.* Type I tube benders are primarily used for making precision bends on cold-drawn corrosion-resisting steel tubing used in aircraft hydraulic systems but can be used for bending many other kinds of metal tubing. Tubing with rough outside surfaces should not be used, as the sheave wheels or blocks may become marred. The radius of a bend on cold-drawn steel tubing should not be less than the minimum specified in table I, as the outer surface of the tubing may be stretched beyond the allowable elongation of the material.

6.1.2 *Type II.* Type II, class 1 tube benders are used for making accurate bends on any tubing except 1-inch and 1-1/8-inch cold-drawn steel tubing. Type II, class 2 tube benders are used for making accurate bends on any tubing except hard-drawn copper or cold-drawn steel tubing having a wall thickness of 0.050 inch or less.

6.1.3 *Type III.* Type III tube benders are used for making accurate bends on any soft tubing. When specified (3.14.1 and 3.14.2), type III benders are also available for bending hard-drawn copper, annealed corrosion-resistant steel, and medium-heavy steel tubing. Such benders, however, require considerable effort or extra long handles when bending such tubing in the larger sizes.

6.1.4 *Type IV.* Type IV tube benders are used for bending soft copper and aluminum tubing to various shapes and as an aid in preventing the tubing from collapsing in making nonprecision bends.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and class required (1.2).

(*c*) Bender size required (tables III, IV, V or VI). For sets of type IV bend-

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.

ers, see 3.15.3 (tables I and II, not applicable).

(*d*) If the bender will be required to bend hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy steel tubing in addition to soft-metal tubing (3.13.2, 3.14.1, and 3.14.2).

(*e*) If a set of type IV benders is required, specify whether the set shall consist of 4 sizes or 6 sizes (3.15.3).

(*f*) If a chest is required for sets of 4 or 6 type IV benders (3.15.4).

(*g*) Level of packaging and packing required (5.1 and 5.2).

6.2.1 When ordering repair parts, specify the following:

(*a*) The manufacturer's name, or trademark of the tool for which the parts are intended.

(*b*) Type, class, and size of tool, if applicable.

(*c*) The name or description of the part, and size if applicable.

(*d*) Quantity of each part required.

6.3 **Transportation description.** The applicable transportation description is as follows:

Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.
Mechanics hand tools, not otherwise indexed.
Truckload weight 30,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—Q

Navy—Sh

Air Force—MOA

Copies of this specification may be purchased for 15 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.